

SHT30 温湿度传感器部分翻译资料

3 Pin Assignment

3、引脚分配

The SHT3x-DIS comes in a tiny 8-pin DFN package see Table 6.

SHT3x-DIS 是一个 8 位的 DFN 小包，见表 6

Pin	Name	Comments
1	SDA	Serial data; input / output
2	ADDR	Address pin; input; connect to either logic high or low, do not leave floating
3	ALERT	Indicates alarm condition; output; must be left floating if unused
4	SCL	Serial clock; input / output
5	VDD	Supply voltage; input
6	nRESET	Reset pin active low; input; if not used it is recommend to be left floating
7	R	No electrical function; to be connected to VSS
8	VSS	Ground

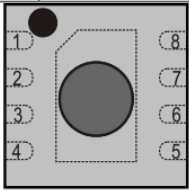


Table 6 SHT3x-DIS pin assignment (transparent top view). Dashed lines are only visible if viewed from below. The die pad is internally connected to VSS.

3.1 Power Pins (VDD, VSS)

3.1、电源引脚（VDD, VSS）

The electrical specifications of the SHT3x-DIS are shown in Table 3. The power supply pins must be decoupled with a 100 nF capacitor that shall be placed as close to the sensor as possible – see Figure 11 for a typical application circuit.

SHT3x-DIS 的电气参数如表 3 所示。电源引脚必须与一个 100nF 的电容器尽可能地隔离，以便尽可能靠近传感器——请参见典型应用电路图 11。

3.2 Serial Clock and Serial Data (SCL, SDA)

3.2、串行时钟和串行数据（SCL, SDA）

SCL is used to synchronize the communication between microcontroller and the sensor. The clock frequency can be freely chosen between 0 to 1000 kHz. Commands with clock stretching according to I2C Standard¹¹ are supported.

SCL 用于同步微控制器和传感器之间的通信。时钟频率可以在 0 到 1000 千赫之间自由选择。支持按 I2C 标准¹¹ 进行时钟扩展的命令。

The SDA pin is used to transfer data to and from the sensor. Communication with frequencies up to 400 kHz must meet the I2C Fast Mode¹¹ standard. Communication frequencies up to 1 Mhz are supported following the specifications given in Table 20.

SDA 引脚用于向传感器传递数据和从传感器传递数据。频率高达 400 kHz 的通信必须满足

I2C 快速模式¹¹ 标准。根据表 20 给出的规格，支持 1 兆赫的通信频率。

Both SCL and SDA lines are open-drain I/Os with diodes to VDD and VSS. They should be connected to external pull-up resistors (please refer to Figure 11). A device on the I2C bus must only drive a line to ground. The external pull-up resistors (e.g. $R_p = 10\text{ k}\Omega$) are required to pull the signal high. For dimensioning resistor sizes please take bus capacity and communication frequency into account (see for example Section 7.1 of NXP's I2C Manual for more details¹¹). It should be noted that pull-up resistors may be included in I/O circuits of microcontrollers. It is recommended to wire the sensor according to the application circuit as shown in Figure 11.

SCL 和 SDA 线均为开路漏极，有二极管连到 VDD 和 VSS。它们应该与外部上拉电阻相连接（请参见图 11）。I2C 总线上的设备必须只有一条线接地。外部上拉电阻（如 $R_p = 10\text{ k}\Omega$ ）需要高的信号。关于尺寸标注的电阻，请将线路容量和通信频率考虑在内（更多细节请参见 NXP's I2C 手册第 7.1 节）。需要注意的是，在微控制器的 I/O 电路中可以包括上拉电阻。建议根据图 11 应用电路连接传感器。

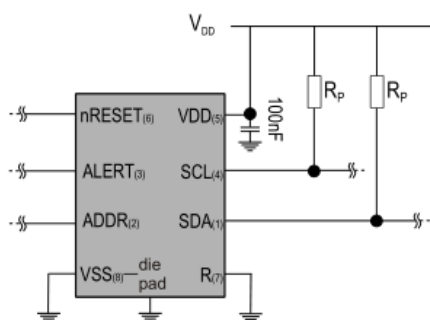


Figure 11 Typical application circuit. Please note that the positioning of the pins does not reflect the position on the real sensor. This is shown in Table 6.

3.3 Die Pad (center pad)

3.3、模具衬垫或中心衬垫

The die pad or center pad is visible from below and located in the center of the package. It is electrically connected to VSS. Hence electrical considerations do not impose constraints on the wiring of the die pad. However, due to mechanical reasons it is recommended to solder the center pad to the PCB. For more information on design-in, please refer to the document “SHTxx Design Guide”.

从下面可见位于小包中心的模具衬垫或中心衬垫。它与 VSS 相连。因此，电气方面的考虑不会对接线板接线造成限制。然而，由于机械原因，建议将焊盘焊到 PCB。关于设计中更多的信息，请参阅文档“SHTxx 设计指南”。

3.4 ADDR Pin

3.4、ADDR 引脚

Through the appropriate wiring of the ADDR pin the I2C address can be selected (see Table 7 for the respective addresses). The ADDR pin can either be connected to VDD or VSS, or it can be used as a selector pin. This means that the address of the sensor can be changed dynamically

during operation by switching the level on the ADDR pin. The only constraint is that the level has to stay constant starting from the I2C start condition until the communication is finished. This allows to connect more than two SHT3x-DIS onto the same bus. The dynamical switching requires individual ADDR lines to the sensors.

通过 ADDR 引脚合适的接线，可以选择 I2C 地址（见表 7 中的相应地址）。ADDR 引脚可以直接连接到 VDD 或 VSS，或者也可以作为一个选择销。这意味着传感器的地址可以改变，在运行过程中动态切换水平上的地址引脚。唯一的限制是电平必须保持恒定，从 I2C 开始状态直到通信结束。这允许将超过两 sht3x DIS 连接在同一总线。动态切换要求单个地址线传感器。

Please note that the I2C address is represented through the 7 MSBs of the I2C read or write header. The LSB switches between read or write header. The wiring for the default address is shown in Table 7 and Figure 11. The ADDR pin must not be left floating. Please note that only the 7 MSBs of the I2C Read/Write header constitute the I2C Address.

请注意，I2C 地址是通过 I2C 读写头的 7 位表达出来的。LSB 在读写头之间切换。默认地址的接线如表 7 和图 11 所示。addr 引脚不能悬空。请注意，只有 I2C 读写头的 7 位构成了 I2C 地址。

SHT3x-DIS	I2C Address in Hex. representation	Condition
I2C address A	0x44 (default)	ADDR (pin 2) connected to VSS
I2C address B	0x45	ADDR (pin 2) connected to VDD

Table 7 I2C device addresses.

3.5 ALERT Pin

3.5、警报引脚

The alert pin may be used to connect to the interrupt pin of a microcontroller. The output of the pin depends on the value of the RH/T reading relative to programmable limits. Its function is explained in a separate application note. If not used, this pin must be left floating. The pin switches high, when alert conditions are met. The maximum driving loads are listed in Table 3. Be aware that self-heating might occur, depending on the amount of current that flows. Self-heating can be prevented if the Alert Pin is only used to switch a transistor.

警报引脚可用于连接微控制器的中断引脚。引脚的输出取决于相对于可编程限制的 RH/ T 读数的值。它的功能会在单独的应用说明中解释。如果不使用，该引脚必须悬空。当警报状态满足时，引脚开关调高。最大驱动载荷列于表 3 中。请注意，可能会发生自加热，这取决于流过的电流。如果警报引脚仅用于开关晶体管，则可以防止自加热。

3.6 nRESET Pin

3.6、NRESET 引脚

The nReset pin may be used to generate a reset of the sensor. A minimum pulse duration of 1 μ s is required to reliably trigger a reset of the sensor. Its function is explained in more detail in section 4.

If not used it is recommended to leave the pin floating.

NRESET 引脚可用于传感器复位。至少 1 μ 的脉冲持续时间需要可靠地触发传感器重置。第 4 节对其功能作了更详细的解释。如果不使用，建议让引脚悬空。

4 Operation and Communication

4、操作和通信

The SHT3x-DIS supports I2C fast mode (and frequencies up to 1000 kHz). Clock stretching can be enabled and disabled through the appropriate user command. For detailed information on the I2C protocol, refer to NXP I2C-bus specification ¹².

SHT3x-DIS 支持 I2C 快速模式（频率高达 1000 kHz）。可以通过适当的用户命令启用或禁用时钟扩展。关于 I2C 协议的详细信息，参见 NXP I2C 总线规范 ¹²。

All SHT3x-DIS commands and data are mapped to a 16-bit address space. Additionally, data and commands are protected with a CRC checksum. This increases communication reliability. The 16 bits commands to the sensor already include a 3 bit CRC checksum. Data sent from and received by the sensor is always succeeded by an 8 bit CRC.

所有 SHT3x-DIS 的命令和数据映射到 16 位地址空间。此外，数据和命令用 CRC 校验和进行保护。这增加了通信的可靠性。传感器的 16 位命令已经包含一个 3 位 CRC 校验和。传感器发送和接收的数据总是由 8 位 CRC 校验成功。

In write direction it is mandatory to transmit the checksum, since the SHT3x-DIS only accepts data if it is followed by the correct checksum. In read direction it is left to the master to read and process the checksum.

在写取方向上必须发送校验和，因为如果后续有正确的校验和，SHT3x-DIS 只接受数据。在读取方向上，留给主控制器读取和处理校验和。

4.1 Power-Up and Communication Start

4.1、启动和通信启动

The sensor starts powering-up after reaching the power-up threshold voltage V_{POR} specified in Table 3. After reaching this threshold voltage the sensor needs the time t_{PU} to enter idle state. Once the idle state is entered it is ready to receive commands from the master (microcontroller). 传感器在达到表 3 中规定的阈值电压后会达到启动功率。达到这个阈值电压后，传感器需要时间 t 进入空闲状态。一旦进入空闲状态，它就可以接收来自主控制器（微控制器）的命令。

Each transmission sequence begins with a START condition (S) and ends with a STOP condition (P) as described in the I2C-bus specification. The stop condition is optional. Whenever the sensor is powered up, but not performing a measurement or communicating, it automatically enters idle state for energy saving. This idle state cannot be controlled by the user.

每个传输序列以起始条件开始，以 I2C 总线规范中描述的停止条件（p）结束。停止条件是可选的。当传感器接通电，但不进行测量或通信时，它就自动进入空闲状态以节约能源。此空闲状态不能由用户控制。

4.2 Starting a Measurement

4.2、开始测量

A measurement communication sequence consists of a START condition, the I2C write header (7-bit I2C device address plus 0 as the write bit) and a 16-bit measurement command. The proper reception of each byte is indicated by the sensor. It pulls the SDA pin low (ACK bit) after the falling edge of the 8th SCL clock to indicate the reception. A complete measurement cycle is depicted in Table 8.

测量通信序列由一个 START 条件，I2C 写头（7 位 I2C 设备地址加 0 作为写取位）和一个 16 位的测量命令组成。传感器会显示每个字节的正确接收。它将 SDA 引脚（ACK 位）拉倒低于第八 SCL 时钟的下降沿，以显示接收。完整的测量周期如表 8 所示。

With the acknowledgement of the measurement command, the SHT3x-DIS starts measuring humidity and temperature.

随着测量命令的确认，SHT3x-DIS 就开始测量温度和湿度。

4.3 Measurement Commands for Single Shot Data Acquisition Mode

4.3、单镜头数据采集模式的测量命令

In this mode one issued measurement command triggers the acquisition of one data pair. Each data pair consists of one 16 bit temperature and one 16 bit humidity value (in this order). During transmission each data value is always followed by a CRC checksum, see section 4.4.

在这种模式下，一个发出的测量命令触发一个数据对的采集。每个数据对由一个 16 位温度和一个 16 位湿度值（按此顺序）组成。在传输过程中，每个数据值总是紧跟着一个 CRC 校验和，见第 4.4 节。

In single shot mode different measurement commands can be selected. The 16 bit commands are shown in Table 8. They differ with respect to repeatability (low, medium and high) and clock stretching (enabled or disabled).

在单镜头模式下，可以选择不同的测量命令。16 位命令如表 8 所示。它们与可重复性（低、中、高）和时钟拉伸（启用或禁用）不同。

The repeatability setting influences the measurement duration and thus the overall energy consumption of the sensor. This is explained in section 2.

重复性设置影响测量持续时间，从而影响传感器的整体能量消耗。这在第 2 节中作了解释。

The I2C master can abort the read transfer with a NACK condition after any data byte if it is not interested in subsequent data, e.g. the CRC byte or the second measurement result, in order to save time.

如果对后续数据（例如 CRC 字节或第二测量结果）不感兴趣，为了节省时间 I2C 主机可以在任何数据字节后使用一个 NACK 条件中止读取传输。

In case the user needs humidity and temperature data but does not want to process CRC data, it is recommended to read the two temperature bytes of data with the CRC byte (without processing the CRC data); after having read the two humidity bytes, the read transfer can be aborted with a NACK.

如果用户需要的温度和湿度数据，但不想处理 CRC 数据，建议读取 CRC 字节数据的两温度字节（无处理 CRC 数据）；在读取两温度字节后，使用一个 NACK 可以终止读取转换。

No Clock Stretching

没有时钟延伸

When a command without clock stretching has been issued, the sensor responds to a read header with a not acknowledge (NACK), if no data is present.

当发出一个没有时钟拉伸的命令时，如果没有数据存在，传感器就会对读取头反应一个不确认（NACK）。

Clock Stretching

时钟延伸

When a command with clock stretching has been issued, the sensor responds to a read header with an ACK and subsequently pulls down the SCL line. The SCL line is pulled down until the measurement is complete. As soon as the measurement is complete, the sensor releases the SCL line and sends the measurement results.

当发出一个时钟拉伸命令时，传感器就会对读取头反应一个 ACK，并随后拉下 SCL 线。直到测量完成，SCL 线才拉下。一旦测量完成，传感器释放 SCL 线并发送测量结果。

4.5 Measurement Commands for Periodic Data Acquisition Mode

4.5、周期性数据采集模式的测量命令

In this mode one issued measurement command yields a stream of data pairs. Each data pair consists of one 16 bit temperature and one 16 bit humidity value (in this order).

在这种模式下，一个发出的测量命令产生一对数据流。每个数据对由一个 16 位温度和一个 16 位湿度值（按此顺序）组成。

In periodic mode different measurement commands can be selected. The corresponding 16 bit commands are shown in Table 9. They differ with respect to repeatability (low, medium and high) and data acquisition frequency (0.5, 1, 2, 4 & 10 measurements per second, mps). Clock stretching cannot be selected in this mode.

在周期模式下，可以选择不同的测量命令。相应的 16 位命令如表 9 所示。它们与重复性（低、

中、高）和数据采集频率（测量每秒 0.5，1, 2, 4 和 10，MPS）不同。在这种模式下不能选择时钟沿伸。

The data acquisition frequency and the repeatability setting influences the measurement duration and the current consumption of the sensor. This is explained in section 2 of this datasheet.
数据采集频率和重复性设置影响传感器的测量持续时间和电流消耗。这在本数据表的第 2 节中进行了说明。

If a measurement command is issued, while the sensor is busy with a measurement (measurement durations see Table 4), it is recommended to issue a break command first (see section 4.8). Upon reception of the break command the sensor will finish the ongoing measurement and enter the single shot mode.
如果发出一个测量命令，当传感器忙于测量（测量持续时间见表 4）时，建议先发出一个中断命令（参见第 4.8 节）。当接收到中断命令时，传感器将完成正在进行的测量并进入单镜头模式。

Condition		Hex. code																																																											
Repeatability	mps	MSB	LSB																																																										
High	0.5	0x20	32																																																										
Medium			24																																																										
Low			2F																																																										
High	1	0x21	30																																																										
Medium			26																																																										
Low			2D																																																										
High	2	0x22	36																																																										
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High	4	0x23	34																																																										
Medium			22																																																										
Low			29																																																										
High	10	0x27	37																																																										
Medium			21																																																										
Low			2A																																																										
e.g. 0x2130: 1 high repeatability mps - measurement per second																																																													
<table><tr><td colspan="18">1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18</td></tr><tr><td colspan="2">S</td><td colspan="6">I2C Address</td><td colspan="2">W</td><td colspan="6">Command MSB</td><td colspan="6">Command LSB</td></tr><tr><td colspan="8">I2C write header</td><td colspan="10">16-bit command</td></tr></table>				1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18																		S		I2C Address						W		Command MSB						Command LSB						I2C write header								16-bit command									
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18																																																													
S		I2C Address						W		Command MSB						Command LSB																																													
I2C write header								16-bit command																																																					

Table 9 Measurement commands for periodic data acquisition mode (Clear blocks are controlled by the microcontroller, grey blocks by the sensor). N.B.: At the highest mps setting self-heating of the sensor might occur.

4.6 Readout of Measurement Results for Periodic Mode

4.6、周期模式测量结果的读出

Transmission of the measurement data can be initiated through the fetch data command shown in Table 10. If no measurement data is present the I2C read header is responded with a NACK (Bit 9 in Table 10) and the communication stops. After the read out command fetch data has been issued, the data memory is cleared, i.e. no measurement data is present.

测量数据的传输可以通过表 10 所示的获取数据命令来启动。如果没有测量数据，I2C 读取头会反应出一个 NACK（表 10 中的 9 位），从而停止通信。在读出命令获取数据之后，数据存储器被清除，即测量数据不会存在。

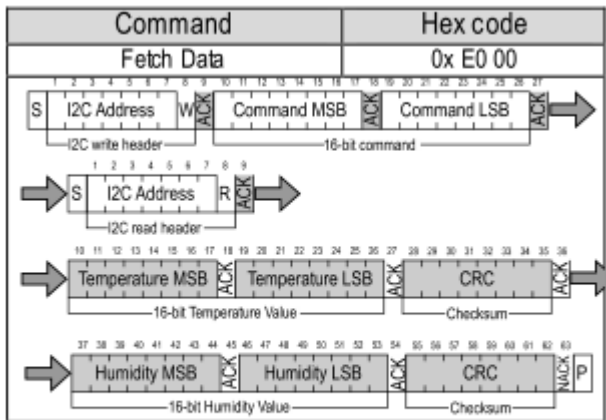


Table 10 Fetch Data command (Clear blocks are controlled by the microcontroller, grey blocks by the sensor).

4.7 ART Command

4.7、ART 命令

The ART (accelerated response time) feature can be activated by issuing the command in Table 11. After issuing the ART command the sensor will start acquiring data with a frequency of 4Hz. ART（加速响应时间）特性可以通过发出表 11 中的命令来激活。发行 ART 命令后，传感器将开始以频率 4Hz 获取数据。

The ART command is structurally similar to any other command in Table 9. Hence section 4.5 applies for starting a measurement, section 4.6 for reading out data and section 4.8 for stopping the periodic data acquisition.

ART 命令在结构上类似于表 9 中的任何其他命令。因此，第 4.5 节适用于启动测量，第 4.6 节适用于读取数据，第 4.8 节适用于停止周期性的数据采集。

The ART feature can also be evaluated using the Evaluation Kit EK-H5 from Sensirion. ART 特征也可以采用 Sensirion 的评估套件 EK-H5 评价。

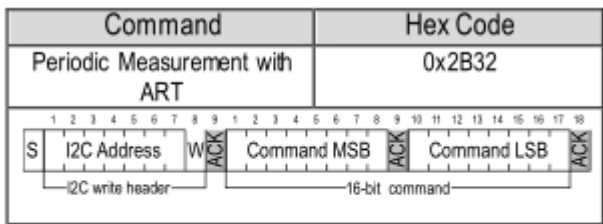


Table 11 Command for a periodic data acquisition with the ART feature (Clear blocks are controlled by the microcontroller, grey blocks by the sensor).

4.8 Break command / Stop Periodic Data Acquisition Mode

4.8、中断命令/停止周期数据采集模式

The periodic data acquisition mode can be stopped using the break command shown in Table 12. It is recommended to stop the periodic data acquisition prior to sending another command (except Fetch Data command) using the break command. Upon reception of the break command the sensor enters the single shot mode, after finishing the ongoing measurement. This can take up to 15 ms, depending on the selected repeatability.

可以使用表 12 所示的中断命令停止周期性数据采集模式。建议使用中断命令发送另一个命令（除了获取数据命令）之前就停止周期性数据获取。在接收到中断命令后，传感器在完成正在进行的测量之后进入单镜头模式。这取决于所选的可重复性，需要多达 15 毫秒。

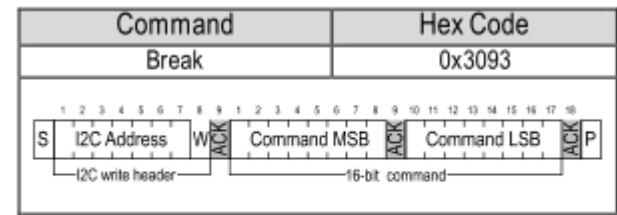


Table 12 Break command (Clear blocks are controlled by the microcontroller, grey blocks by the sensor).

4.9 Reset

4.9 复位

A system reset of the SHT3x-DIS can be generated externally by issuing a command (soft reset) or by sending a pulse to the dedicated reset pin (nReset pin). Additionally, a system reset is generated internally during power-up. During the reset procedure the sensor will not process commands.

可以从外部发出命令（软复位）或通过发送一个脉冲到专用的复位引脚（nRESET 引脚），对 SHT3x-DIS 进行系统复位。此外，系统复位是在电源上升期间产生的。在重置过程中，传感器将不处理命令。

In order to achieve a full reset of the sensor without removing the power supply, it is recommended to use the nRESET pin of the SHT3x-DIS.

为了实现传感器的全复位，无需拆卸电源，建议使用的 SHT3x-DIS 的 nRESET 引脚。

Interface Reset

接口复位

If communication with the device is lost, the following signal sequence will reset the serial interface: While leaving SDA high, toggle SCL nine or more times. This must be followed by a Transmission Start sequence preceding the next command. This sequence resets the interface only. The status register preserves its content.

如果与设备的通信丢失，则以下信号序列将重置串行接口：在离开 SDA 高电平时，切换 SCL

九次或更多次。在下一个命令之前，必须有一个传输开始序列。此序列仅重置接口。状态寄存器将保存其内容。

Soft Reset / Re-Initialization
软复位/重新初始化

The SHT3x-DIS provides a soft reset mechanism that forces the system into a well-defined state without removing the power supply. When the system is in idle state the soft reset command can be sent to the SHT3x-DIS. This triggers the sensor to reset its system controller and reloads calibration data from the memory. In order to start the soft reset procedure the command as shown in Table 13 should be sent.

SHT3x-DIS 提供软复位机制，可以强制系统进入一个明确的状态，而无需拆卸电源。当系统处于空闲状态，软复位命令可以发送到 SHT3x-DIS。这将使传感器复位其系统控制器，并从存储器重新下载校准数据。为了启动软复位程序，应发送表 13 所示的命令。

It is worth noting that the sensor reloads calibration data prior to every measurement by default. 值得注意的是，传感器会默认在每次测量前重新加载标定数据。

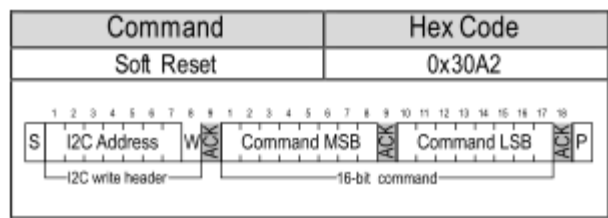


Table 13 Soft reset command (Clear blocks are controlled by the microcontroller, grey blocks by the sensor).

Reset through General Call
通过通用呼叫进行重置

Additionally, a reset of the sensor can also be generated using the “general call” mode according to I2C-bus specification 12 . This generates a reset which is functionally identical to using the nReset pin. It is important to understand that a reset generated in this way is not device specific. All devices on the same I2C bus that support the general call mode will perform a reset. Additionally, this command only works when the sensor is able to process I2C commands. The appropriate command consists of two bytes and is shown in Table 14.

此外，还可以根据 I2C 总线规范 12 使用“通用呼叫”模式来复位传感器。这样产生的复位与使用 nRESET 引脚一样。重要的是要理解，以这种方式生成的重置不是特定于某个设备的。在支持通用呼叫模式的同一 I2C 总线上的所有设备都将执行复位。此外，只有当传感器能够处理 I2C 命令时，此命令才起作用。适当的命令由两个字节组成，如表 14 所示。

Reset through the nReset Pin
通过 nRESET 引脚复位

Pulling the nReset pin low (see Table 6) generates a reset similar to a hard reset. The nReset pin is internally connected to VDD through a pull-up resistor and hence active low. The nReset pin has to be pulled low for a minimum of 1 μ s to generate a reset of the sensor.

将 nRESET 引脚拉低（见表 6）将产生一个类似硬复位的复位。nRESET 引脚通过上拉电阻，与 VDD 内部连接，因此位于低电平。nReset 引脚要拉低至少 1 μ 才能复位传感器。

Command	Code
Address byte	0x00
Second byte	0x06
Reset command using the general call address	0x0006

Table 14 Reset through the general call address (Clear blocks are controlled by the microcontroller, grey blocks by the sensor).

Hard Reset

硬复位

A hard reset is achieved by switching the supply voltage to the VDD Pin off and then on again. In order to prevent powering the sensor over the ESD diodes, the voltage to pins 1 (SDA), 4 (SCL) and 2 (ADDR) also needs to be removed.

一个硬复位是将电源电压调到 VDD 引脚关闭状态，再调到开启状态，即达到应复位。为了防止供电的传感器超过 ESD 二极管，需要将电压引脚 1（SDA）、4（SCL）和 2（ADDR）移除。

4.10 Heater

4.10、加热器

The heater can be switched on and off by command, see table below. The status is listed in the status register. After a reset the heater is disabled (default condition).

加热器通过命令进行开启和关闭，见下表。状态被列在状态寄存器中。复位后，加热器就被禁用（默认条件）。

Command	Hex Code	
	MSB	LSB
Heater Enable	0x30	6D
Heater Disabled		66

The timing diagram illustrates the sequence of events for the heater command. It shows a sequence of bytes on a bus with pins 1-18. The sequence starts with a Start (S) condition, followed by an I2C Address (7 bytes), a Write (W) condition, then a 16-bit command (split into MSB and LSB, each 8 bytes), and ends with an ACK and a Stop (P) condition. The I2C write header is at pin 1, and the 16-bit command is at pin 2.

Table 15 Heater command (Clear blocks are controlled by the microcontroller, grey blocks by the sensor).

4.11 Status Register

4.11、状态登记

The status register contains information on the operational status of the heater, the alert mode and on the execution status of the last command and the last write sequence. The command to read out the status register is shown in Table 16 whereas a description of the content can be found in Table 17.

状态寄存器包含加热器的操作状态、警报模式以及上一个命令和最后一个写入序列的执行状态的信息。读取状态寄存器的命令如表 16 所示，而对内容的描述可在表 17 中找到。

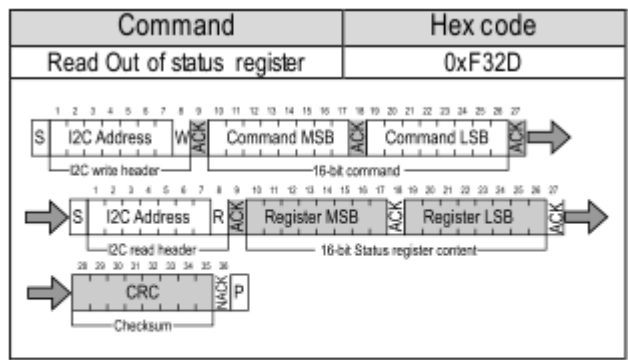


Table 16 Command to read out the status register (Clear blocks are controlled by the microcontroller, grey blocks by the sensor).

Bit	Field description	Default value
15	Alert pending status '0': no pending alerts '1': at least one pending alert	'1'
14	Reserved	'0'
13	Heater status '0': Heater OFF '1': Heater ON	'0'
12	Reserved	'0'
11	RH tracking alert '0': no alert '1': alert	'0'
10	T tracking alert '0': no alert '1': alert	'0'
9:5	Reserved	'xxxxx'
4	System reset detected '0': no reset detected since last 'clear status register' command '1': reset detected (hard reset, soft reset command or supply fail)	'1'
3:2	Reserved	'00'
1	Command status '0': last command executed successfully '1': last command not processed. It was either invalid, failed the integrated command checksum	'0'
0	Write data checksum status '0': checksum of last write transfer was correct '1': checksum of last write transfer failed	'0'

Table 17 Description of the status register.

Clear Status Register

明确的状态登记

All flags (Bit 15, 11, 10, 4) in the status register can be cleared (set to zero) by sending the command shown in Table 18.

状态寄存器中的所有标志（15, 11, 10, 4 位）可以通过发送表 18 所示的命令进行清除（设置为零）。

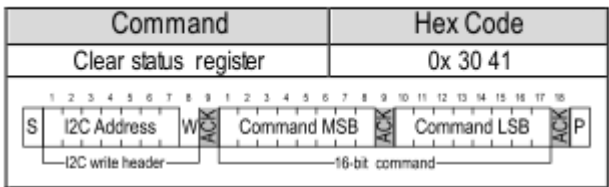


Table 18 Command to clear the status register (Clear blocks are controlled by the microcontroller, grey blocks by the sensor).

4.12 Checksum Calculation

4.12、校验和计算

The 8-bit CRC checksum transmitted after each data word is generated by a CRC algorithm. Its properties are displayed in Table 19. The CRC covers the contents of the two previously transmitted data bytes. To calculate the checksum only these two previously transmitted data bytes are used.

在每个数据字之后发送的 8 位 CRC 校验和是由 CRC 算法生成的。它的属性显示在表 19 中。CRC 覆盖了先前传输的两个数据字节的内容。要计算校验和，只使用这两个先前发送的数据字节即可。

Property	Value
Name	CRC-8
Width	8 bit
Protected data	read and/or write data
Polynomial	0x31 ($x^8 + x^5 + x^4 + 1$)
Initialization	0xFF
Reflect input	False
Reflect output	False
Final XOR	0x00
Examples	CRC (0xBEEF) = 0x92

Table 19 I2C CRC properties.

4.13 Conversion of Signal Output

4.13、信号输出转换

Measurement data is always transferred as 16-bit values (unsigned integer). These values are already linearized and compensated for temperature and supply voltage effects. Converting those raw values into a physical scale can be achieved using the following formulas. Relative humidity conversion formula (result in %RH):

测量数据总是以 16 位值（无符号整数）传输。这些值已经线性化并补偿了温度和电源电压的影响。使用以下公式可以将这些原始值转换为物理量。相对湿度换算公式为（结果为% RH）:

$$RH = 100 \cdot \frac{S_{RH}}{2^{16} - 1}$$

Temperature conversion formula (result in °C & °F):

温度换算公式为（结果为 °C & °F）:

$$T [^{\circ}\text{C}] = -45 + 175 \cdot \frac{S_T}{2^{16} - 1}$$

$$T [^{\circ}\text{F}] = -49 + 315 \cdot \frac{S_T}{2^{16} - 1}$$

S_{RH} and S_T denote the raw sensor output for humidity and temperature, respectively. The formulas work only correctly when S_{RH} and S_T are used in decimal representation.

S_{RH} 和 S_T 分别表示原传感器输出的温度和湿度。只有当 S_{RH} 和 S_T 用十进制表示时，该公式才能计算正确。

4.14 Communication Timing

4.14、通信时序

Parameter	Symbol	Conditions	Min.	Typ.	Max.	Units	Comments
SCL clock frequency	f _{SCL}		0	-	1000	kHz	
Hold time (repeated) START condition	t _{HD,STA}	After this period, the first clock pulse is generated	0.24	-	-	μs	
LOW period of the SCL clock	t _{LOW}		0.65	-	-	μs	
HIGH period of the SCL clock	t _{HIGH}		0.26	-	-	μs	
SDA hold time	t _{HD,DAT}		0	-	250	ns	Transmitting data
			0	-	-	ns	Receiving data
SDA set-up time	t _{SU,DAT}		100	-	-	ns	
SCL/SDA rise time	t _r		-	-	300	ns	
SCL/SDA fall time	t _f		-	-	300	ns	
SDA valid time	t _{VD,DAT}		-	-	0.9	μs	
Set-up time for a repeated START condition	t _{SU,STA}		0.6	-	-	μs	
Set-up time for STOP condition	t _{SU,STO}		0.6	-	-	μs	
Capacitive load on bus line	CB		-	-	400	pF	
Low level input voltage	V _{IL}		-0.5	-	0.3xV _{DD}	V	
High level input voltage	V _{IH}		0.7xV _{DD}	-	1xV _{DD}	V	
Low level output voltage	V _{OL}	3 mA sink current	-	-	0.66	V	

Table 20 Communication timing specifications for I2C fm (fast mode), specifications are at 25°C and typical VDD. The numbers above are values according to the I2C Specification (UM10204, Rev. 6, April 4, 2014).

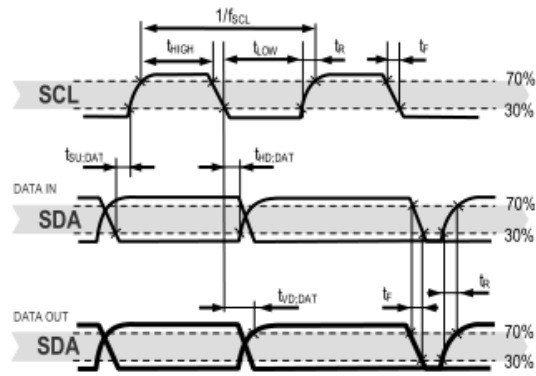


Figure 12 Timing diagram for digital input/output pads. SDA directions are seen from the sensor. Bold SDA lines are controlled by the sensor, plain SDA lines are controlled by the micro-controller. Note that SDA valid read time is triggered by falling edge of preceding toggle.